Tic Tac Game using MIN MAX

```python
board = [" " for i in range(9)]

def display():
    for i in range(0, 9, 3):
        print(board[i], "|", board[i+1], "|", board[i+2])
        if i < 6:
            print("---------")

def check_winner(player):
    win = [
        [0,1,2], [3,4,5], [6,7,8],
        [0,3,6], [1,4,7], [2,5,8],
        [0,4,8], [2,4,6]
    ]

    for a, b, c in win:
        if board[a] == board[b] == board[c] == player:
            return True
    return False

def minimax(is_max):
    if check_winner("O"):
        return 1
    if check_winner("X"):
        return -1
    if " " not in board:
        return 0

    if is_max:
        best = -999
```

```
        for i in range(9):
            if board[i] == " ":
                board[i] = "O"
                score = minimax(False)
                board[i] = " "
                best = max(best, score)

        return best

    else:
        best = 999

        for i in range(9):
            if board[i] == " ":
                board[i] = "X"
                score = minimax(True)
                board[i] = " "
                best = min(best, score)

        return best

def computer_move():
    best = -999
    move = -1

    for i in range(9):
        if board[i] == " ":
            board[i] = "O"
            score = minimax(False)
            board[i] = " "
```

```
while True:
    display()
    row, col = map(int, input("Enter row and column (0-2): ").split())
    pos = row * 3 + col

    if row < 0 or row > 2 or col < 0 or col > 2 or board[pos] != " ":
        print("Invalid move! Try again.")
        continue

    board[pos] = "X"

    if check_winner("X"):
        display()
        print("Human Player Wins")
        break

    if " " not in board:
        display()
        print("Game Draw")
        break

    computer_move()

    if check_winner("O"):
        display()
        print("Computer Wins")
        break

    if " " not in board:
        display()
        print("Game Draw")
```

Output:

```
TIC TAC TOE
You are X and Computer is O
  |   |
---------
  |   |
---------
  |   |
Enter row and column (0-2): 0 0
X |   |
---------
  | O |
---------
  |   |
Enter row and column (0-2): 0 1
X | X | O
---------
  | O |
---------
  |   |
Enter row and column (0-2): 1 0
X | X | O
---------
X | O |
---------
O |   |
Computer Wins
```

```
TIC TAC TOE
You are X and Computer is O
  |   |
---------
  |   |
---------
  |   |
Enter row and column (0-2): 0 0
X |   |
---------
  | O |
---------
  |   |
Enter row and column (0-2): 0 1
X | X | O
---------
  | O |
---------
  |   |
Enter row and column (0-2): 2 0
X | X | O
---------
O | O |
---------
X |   |
Enter row and column (0-2): 1 2
X | X | O
---------
O | O | X
---------
X | O |
Enter row and column (0-2): 2 2
X | X | O
---------
O | O | X
---------
X | O | X
Game Draw
```